

Checking Alethe Proofs with LEAN-SMT

A summary of the work on the `alethe` branch of LEAN-SMT

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Abstract. LEAN-SMT [2] reconstructs CVC5’s CPC proofs into Lean proofs, driving the solver through its API. This note describes an extension that checks textual Alethe proofs, as produced by CVC5 and VERiT and elaborated by CARCARA, against SMT-LIB problems, reusing LEAN-SMT’s term reconstruction, lemma libraries, and arithmetic decision procedures. We describe the pipeline, how Alethe’s contexts and clauses are handled, how the proof rules are reconstructed, and how the proof reaches the kernel: step by step, in batches, or as one term. We close with the state of the evaluation and what it revealed about the proofs themselves.

1 System Overview

Figure 1 depicts the pipeline. The input is an SMT-LIB problem and an Alethe proof of its unsatisfiability, produced by CVC5 (`--proof-format-mode=alethe`) or by VERiT. The proof first goes through CARCARA’s [3] *elaborator*, which turns the coarse rules of the solver output into steps the checker can replay: the `polyeq` pass removes the polyequalities and implicit contexts of `assume`, `refl` and `forall_inst`; the `local` pass canonicalizes `trans`, `cong` and resolution, writing the pivots as arguments; and the `core` pass reduces the rules we chose not to reconstruct in Lean to rules we do reconstruct, restricted to a list given on the command line. The elaborated proof is then read by LEAN-SMT. Its *parser* eliminates the binding constructs of the text, its *realizer* hands every term to CVC5’s own parser through the Lean API, so that the reconstruction of terms is the one LEAN-SMT already has for CPC proofs, and its *reconstructor* replays the proof rule by rule, sending each step to Lean’s kernel as it goes. The verdict distinguishes a *valid* proof, in which every step was reconstructed and accepted by the kernel, from a *holey* one, in which some steps had to be trusted; the trusted steps are reported by rule, and, when the checker runs as a tactic, become goals for the user.

The checker is available in three forms. The command `#check_alethe "P.smt2"` `"P.alethe"` checks a proof from files and prints the verdict with per-phase timings; the `alethe` tactic closes a Lean goal the way the `smt` tactic does, translating the goal, obtaining an Alethe proof from CVC5 (in process, or from an external binary), running CARCARA, and reconstructing the result as a proof term of the goal; and the underlying `checkAlethe` function serves an executable for batch evaluation. The trusted computing base is the one of CPC reconstruction: Lean’s kernel, and CVC5’s parser for the statement.

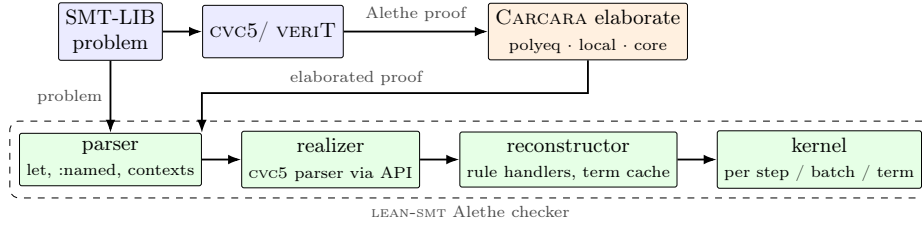


Fig. 1. Checking an Alethe proof. The solver’s proof is elaborated by CARCARA; the checker parses and realizes it with CVC5’s parser, replays it with LEAN-SMT’s reconstruction, and submits the steps to Lean’s kernel.

The rest of this section follows the pipeline: the treatment of the proof text (Section 1.1), the replay of steps and their submission to the kernel (Section 1.2), the reconstruction techniques per rule (Section 1.3), and the changes this required in CARCARA (Section 1.4). Section 2 reports the evaluation, Section 3 the comparison of the kernel submission shapes, and Section 4 that of the **native** variant against the kernel-checked default.

1.1 From Proof Text to Terms

Binding constructs. An Alethe proof, as CARCARA prints it, shares subterms with $(! \ t \ :named \ @p_n)$ annotations and, in VERiT’s case, with **let**. The parser eliminates both over a hash-consed arena of S-expressions, so that substitution never copies a term: a **let** is a scoped parallel binding, and a name stands for the annotated term from its definition onward. Anchor variables, the $(x \ S)$ and $(:= x \ t)$ arguments of an **anchor** command, are renamed to globally fresh symbols inside their block. The realizer then serializes each literal once, binding every shared subterm on first use as a fresh constant $s\sim n$, so that CVC5’s hash-consing and LEAN-SMT’s term cache keep the Lean terms shared as well.

Contexts. Alethe’s subproofs under an anchor with $(:= x \ t)$ arguments carry a context Γ , and a step $\Gamma \triangleright s \simeq u$ inside it means $subst(\Gamma)(s) = u$: the substitution applies to the left-hand side only, and $subst(\Gamma, x \mapsto t) = subst(\Gamma) \circ \{x \mapsto t\}$, as in the semantics of Barbosa et al. for formula processing [1] and in CARCARA. The checker encodes an assignment as a **let**-bound local, so that inside the block the replaced variable unfolds to its value definitionally, and the bind rules read the substitution off that unfolding. To give the two sides of a judgment their two meanings, the parser carries two environments per context, the *substituted* one (with the renamings) and the *fixed* one (the enclosing environment plus the context’s own variables), and parses a unit clause $(= \ 1 \ r)$ with 1 under the first and r under the second. Where no substitution is in scope the two coincide. This is what makes VERiT’s nested renamings checkable, where an inner **bind** replaces a name that the enclosing **bind** had just introduced as a fixed variable; such proofs are valid under the paper’s calculus but outside the canonical fragment its algorithm produces, and were previously unrepresentable.

The same machinery absorbs a quirk of VERiT’s sharing: with **--proof-with-sharing**, VERiT names terms that are *open* in let-bound variables and reuses the names

under other bindings of those variables. Read as SMT-LIB, where a name denotes the closed term at its definition, most of VERiT’s QF_UF proofs state assumptions that are not the problem’s assertions. Read as macros over the let-bound variables they mention, they do. Both CARCARA and the checker now resolve a name anew in every scope it is used in, closed names being unaffected.

Realization. Terms are realized by CVC5’s parser in the higher-order logic HO_ALL, so that a `choice` binder can be represented as a per-sort constant applied to a `lambda` and later mapped to `Classical.epsilon`. Integer numerals in real-only logics are read as reals, as both solvers do. From this point on, the reconstruction of sorts and terms is LEAN-SMT’s `reconstructTerm`, unchanged.

1.2 Replaying Steps and Reaching the Kernel

Steps and clauses. A step is handed to the rule reconstructors as a record of its clause literals, rule, premises (with their literals and the Lean proof of their clause), arguments, discharged assumptions, and, for the closing step of a subproof, the anchor’s variables, assignments and last step. Reconstructors are registered with an attribute, `@[alethe_rule_reconstruct]`, one per theory, and dispatch on the rule name. A clause `(cl l1 ... ln)` is the right-nested disjunction of its literals, definitionally `orN [l1, ..., ln]`, so that the `orN` lemmas of LEAN-SMT apply. Alethe’s clauses are sets, and elaborated resolutions state their conclusions modulo permutation and duplicate removal, so a computed clause is aligned with the stated one by `orN_of_map`: an index map from the stated literals to the computed ones, checked by the kernel by evaluation, proves the stated clause from the computed one; the associative-commutative rewriting of `ac_rfl` is kept only as a fallback.

Step by step. CPC reconstruction in LEAN-SMT builds one proof term for the whole proof and submits it to the kernel once. The Alethe checker instead checks every top-level step on its own: the step’s proof is type-checked by `Kernel.check` against its stated clause in a local context holding the problem’s symbols and assertions and only the earlier steps the proof mentions, as hypotheses; the proof term is then dropped and the step becomes a hypothesis for the steps after it. Steps inside a subproof are kept as terms, since the closing step’s proof abstracts over the block’s assumptions and variables, and are checked with it. Memory stays bounded by the largest step rather than the whole proof, a failure is attributed to the step the kernel rejected, and the checked count is exact. The cost is that every kernel call starts with an empty inference cache.

Batches, and one term. Between the two extremes, consecutive top-level steps can be gathered into one kernel call. A batch $e_1 : c_1, \dots, e_n : c_n$ becomes the term

$$(\lambda h_1 : c_1. (\lambda h_2 : c_2. \dots (\lambda h_n : c_n. \text{trivial}) e_n \dots) e_2) e_1,$$

which the kernel accepts exactly when every e_i proves c_i under the hypotheses of the steps before it: the binders keep those hypotheses opaque, each conclusion stays an explicit annotation, and one call shares its cache over the batch. Each proof is abstracted once over the earlier hypotheses of its batch, so construction

is linear in the batch. The batch is bounded by a number of steps or by the number of term nodes it holds (counted with sharing), and a rejected batch is re-checked step by step, so that what is trusted is what the kernel actually rejected. Batches can also be dispatched to concurrent tasks, at most a fixed number in flight, joined in step order; the calls are independent by construction, since each establishes its conclusions from hypotheses rather than from the earlier proofs. Finally, a *term mode* mirrors CPC reconstruction, inlining every step into the steps that use it and submitting one closed theorem; it is what the **alethe** tactic uses, and the shape the evaluation compares against.

1.3 Reconstructing the Rules

The checker handles 93 Alethe rule names, in about 4 800 lines of Lean plus the generated RARE table, following the three techniques of CPC reconstruction: theorems, tactics, and verified programs. The dividing line with CARCARA is drawn per rule: a rule is reconstructed in Lean when a direct proof is cheap and robust, and reduced by CARCARA’s core pass otherwise, which the **--core-rules** flag we added lets the pipeline select rule by rule. Three reconstructors were retired once the line settled there: **bfun_elim** and **th_resolution**, which the core pass eliminates, and **strict_resolution**, whose set semantics the **resolution** reconstructor already gave. None of the three occurs in any elaborated proof of the corpus, and removing them left every verdict unchanged.

Via theorems. The clausal rules of Alethe, the CNF axioms (**and_pos**, **or_neg**, **implies_pos**, **equiv_neg1**, the **ite** and **xor** families) and the elimination rules (**and**, **not_or**, **implies**, **equiv2**, ...), are the CPC lemmas of LEAN-SMT applied with the indices CARCARA’s **local** pass writes as arguments. The connective definitions, the selection axioms **ite_then_intro** and **ite_else_intro** that the core pass derives **ite_intro** from, **la_disequality**, the division rules of CVC5 (**div_intro**, **div_by_zero_intro**) and the like are new lemmas in an Alethe-specific library of 52 theorems.

Via tactics. **la_generic** is Farkas’ lemma: the negated literals are introduced as hypotheses, integer bounds are tightened by the gcd of their coefficients as CARCARA does (a strict bound with unit gcd by **Int.add_one_le_iff** alone; the rounding cases by **omega**, given just the bound, the shape **poly_norm** establishes and the negated goal), the bounds are scaled by the rational coefficients given as arguments (integer bounds cast to rationals in mixed combinations), summed with LEAN-SMT’s **sum_ub** lemmas, normalized by **poly_norm**, and closed by the sign of the constant. Both details matter at scale: a **la_generic** with five hundred literals ran five hundred **omega** calls over the whole local context before, and now takes a few seconds. **cong** aligns the premises with the argument positions, in either orientation, and falls back to definitional equality for arguments that differ only by the context’s substitutions or by **Decidable** instances; congruence over n -ary **distinct**, whose reconstruction is a conjunction rather than an application, rewrites the arguments one at a time. **onepoint** is proved directly as a propositional extensionality: the eliminated binder may sit anywhere in the prefix and the guard may occur under further binders. The

binder rules (`bind`, `sco_forall`, `sco_ex`, `let`) abstract the block's variables with `forall_congr`, `exists_congr_eq` and the epsilon lemmas of LEAN-SMT; the clausal form of `bind` closes the quantified literal wherever it sits in the clause.

Via reflection. `poly_simp` and `poly_simp_rel` reuse LEAN-SMT's verified polynomial normalizer, and `evaluate` the `decide` path. The normalizer proves an equation by `Eq.refl` on the normal forms of its two sides, so the kernel is what computes them, and its cost on the thousand-variable sums of the SMPT benchmarks was the largest single source of failures in the evaluation. It folded the expression through a sorted insertion, copying the partial polynomial at every term of a left-associated sum, which is what SMT-LIB's n -ary `+` reconstructs to: a sum of 400 variables took the kernel 25 s and 8.9 GB, and a thousand exhausted the memory limit. It now does what CARCARA does, transposed to what the kernel can reduce: the monomials are collected in one pass into a flat list (carrying a sign, or a rational multiplier, so that subtraction and division by a constant cost nothing) and sorted once by a bottom-up merge sort whose merge adds like monomials. Everything stays structurally recursive, with the sort's pass count a `Nat` bound, because the kernel does not unfold well-founded recursion; the denotation theorems hold for unsorted input too, so sortedness is needed only for completeness. The same sum of 400 variables now costs the kernel 1.3 s, and 800 cost 3.6 s at flat memory. The associative-commutative rules are the other reflective family. Alethe's `aci_simp` named one normal form for a small algebraic hierarchy of operators, with `absorb` for the annihilator on the side; the rules are now named by the structure of the operator (Section 1.4), and each has the normalizer of exactly that structure. `semilattice_simp` covers the bounded semilattices $\wedge$ and $\vee$: associativity, commutativity, idempotence, the neutral element and the annihilator. Associative-commutative rewriting took seconds per step on VERIT proofs with two-hundred-conjunct layers, so we wrote a normalizer in the style of `poly_norm`:

```

inductive Tree | unit | absorb | leaf (i : Nat) | node (l r : Tree)
def bits : Tree → Option Nat
| unit => some 0
| absorb => none
| leaf i => some (Nat.shiftLeft 1 i)
| node l r => match bits l, bits r with
| some a, some b => some (Nat.lor a b)
| _, _ => none
theorem denoteAnd_eq (ctx : Context) (a b : Tree) (m : Option Nat)
(ha : bits a = m) (hb : bits b = m) :
denoteAnd ctx a = denoteAnd ctx b

```

The top layer of a side is reified into a `Tree` whose leaves index a context of atoms, every subterm that is not the connective or one of its two constants being an atom compared syntactically, as in CARCARA's checker; its normal form is the bitset of its leaves, `none` when the annihilator occurs, which the kernel computes in one structural pass with its accelerated `Nat.shiftLeft` and `Nat.lor`, and `denoteAnd_eq` turns two equal normal forms, both established by

`Eq.refl`, into the equality of the formulas. Soundness is `denoteAnd ctx t ↔ ∀ i, testBit m i → ctx i` when `bits t = some m`, and `denoteAnd ctx t ↔ False` when it is `none`, both by induction on the tree. A 200-atom layer costs the kernel 33 ms, against 34 s for a first attempt with a nested inductive and an insertion sort. `boolean_group_simp` covers `xor`, an abelian group of exponent two, with the same machinery and a different fold: the bitset is combined with `Nat.xor` instead of `Nat.lor`, so that a pair $x \oplus x$ cancels and the normal form is the *parity* of the atoms; soundness goes through the parity of the set bits of a mask over the bit positions, with the interchange law $(a \oplus b) \oplus (c \oplus d) = (a \oplus c) \oplus (b \oplus d)$ doing the regrouping. The two are siblings, not one rule: idempotence keeps a bit and self-inverse toggles it. `assoc_simp`, for the concatenations, has no operator in the checker’s scope. The semilattice normalizer also serves CVC5’s `ACI_NORM` in CPC proofs.

Rewrite rules. CVC5’s rewrites reach Alethe as `rare_rewrite` steps naming a RARE rule and its arguments. A generator reads the RARE file `CARCARA` uses and produces, per rule, a Lean theorem statement and an entry of a name-keyed table; the proofs are written by hand between markers the generator preserves. 117 theorems cover the 107 rules the table lists, with `Int` and `Rat` instances chosen by the arguments’ sorts, list arguments spliced, and integer arguments cast where a rule mixes sorts; five statements of the multiplication-tangent family are stated but not yet proved, and seven n -ary rules have no statement, one of which, `distinct-false`, is handled by a dedicated proof from the reflexive disequality of the expanded conjunction. The `*_simplify` rules of Alethe are not reconstructed at all: the core pass, restricted to them with `--core-rules`, rewrites each into a chain of RARE steps, which the table then covers.

Legacy rules. VERIT’s `ac_simp`, `qnt_cnf`, `ite_intro` and `bfun_elim` are likewise delegated to the core pass, which reduces `bfun_elim` completely, both the expansion of a quantifier over Boolean variables into its instances and the replacement of a Boolean argument of an application by an if-then-else, derives `ite_intro` from the selection axioms, and decomposes a nested `ac_simp` into one `semilattice_simp` per connective layer lifted by `cong` — the single-layer normalizer sees a nested opposite connective as an atom, so the decomposition is what lets the reflective check apply layer by layer. `aci_simp` and `absorb` themselves are relabeled by the core pass to the structural rule of their operator, so the checker reconstructs only `poly_simp` and the three structural rules. The application form of `bfun_elim` is also reconstructed natively, by case analysis on the Boolean arguments, for proofs elaborated without delegating it.

Instances. One consequence of substitution is worth recording. The instantiation of a quantified Boolean variable that occurs in an if-then-else condition carries the `Decidable` instance the quantified body was reconstructed with, `Classical.propDecidable` for an open condition, while the stated conclusion, reconstructed on its own, carries the structural instance for the now-closed condition. The two are propositionally but not definitionally equal, so the kernel rejects the step. A small procedure walks the two terms and closes such dif-

ferences with `Subsingleton.elim`; `forall_inst` applies it when the instantiated body and the stated conclusion are not definitionally equal.

1.4 Changes in CARCARA

Besides being the elaborator, CARCARA is the reference the checker is measured against, and this work changed it in several places, all on the `bv-fixes` branch. The `--core-rules` flag restricts the core passes to the listed rules, so that a pipeline can delegate exactly the rules its consumer does not reconstruct. Under `--expand-let-bindings`, the `let` steps and their `bind_let` subproofs are trivialized, since a consumer expanding lets has nothing left to prove there. The `polyeq` pass canonicalizes Skolem witnesses, and the `sko_forall` and `sko_ex` rules become strict under elaborated granularity, which removed the last discrepancy in VERIT’s nested Skolemizations; the same pass replaces every `bind` subproof whose two sides are alpha-equivalent — VERIT’s identity substitutions and pure renamings — by one `refl`, which halves the `bind` subproofs of a typical Sledgehammer proof; `qnt_join` merges a whole prefix of nested quantifiers. The same pass also canonicalizes `distinct_elim`: it builds the elimination itself, the pairs in index order and each conjunct oriented $\neg(x_i = x_j)$ with $i < j$, and bridges from there to whatever the solver wrote, so the consumer never sees a permuted or flipped conjunct. That one case is worth 5.1 CPU hours of the round (Section 2.2). Named terms open in let-bound variables are macros under let expansion, as described in Section 1.1.

A `budget` pass splits a `resolution` over more than `--resolution-budget` premises (64 by default) into a chain of shorter resolutions with the contraction and weakening steps the pieces need; it turned the two chains of a QF_UF proof the kernel could not hold in memory into pieces of a few seconds each, at 90 extra steps in 40 000.

The associative-commutative vocabulary was reorganized. `aci_simp` normalized operators that form a hierarchy — the bounded semilattices `and`, `or`, `bvand`, `bvor` (associative, commutative, idempotent, with a unit and an annihilator), the abelian groups of exponent two `xor` and `bvxor` (every element its own inverse), the free monoids `concat` and `str.++` (not commutative), and the commutative rings `+`, `*`, `bvadd`, `bvmul` — under one name and one normal form, and `absorb` stated the annihilator law on its own. Three checker rules now split them by structure, each with exactly its operators’ normal form: `semilattice_simp` (a set, collapsing to the annihilator when it occurs), `boolean_group_simp` (a parity) and `assoc_simp` (a sequence); the rings are `poly_simp`’s, whose polynomial normal form also covers the finite-field operators `ff.add` and `ff.mul` modulo the characteristic. The core pass relabels `aci_simp` and `absorb` to the structural rule of their operator after that rule’s checker accepts the conclusion, and emits the layers of `ac_simp`, `shuffle` and `nary_elim` as structural rules, so no legacy AC name survives it; the legacy checkers remain for proofs checked without elaboration. Finally, the core recipe for `bool_simplify`’s modus-ponens rewrites chose the implication conjunct by shape and failed when the antecedent was itself an implication; it now takes it from the rewrite’s label. The core transformation of `bfun_elim` and `ite_intro` was revised on the `coreAlethe` branch and merged.

2 Evaluation

Two corpora, and on each the four configurations that matter for the question at hand: each solver’s proof checked by CARCARA, and elaborated by CARCARA and checked by LEAN-SMT. The LEAN-SMT arm is the per-step shape, the default (Section 1.2); the other shapes are compared in Section 3. Every benchmark runs on one core of a 16-core node with 12 GB and 3 000 s: 150 s for the solver, 300 s each for CARCARA’s check of the solver’s proof, the elaboration, and CARCARA’s check of the *elaborated* proof, and 900 s for the checker. That third CARCARA arm carries no result of its own; it exists so that Section 2.2 can compare the two checkers rule by rule over the same steps. Times below are CPU, not wall: the checker reads its build over a network filesystem, which costs several seconds of wall time on a cold node.

2.1 SMT-LIB

The corpus. The unsatisfiable non-incremental SMT-LIB benchmarks of the twelve logics in the checker’s scope — QF_UF, QF_IDL, QF_RDL, QF_LIA, QF_LRA, QF_UFIDL, QF_UFLIA, QF_UFLRA, UF, UFIDL, UFLIA and UFLRA — as the SMT-LIB 2025 catalog lists them: 23 328 per solver.

Outcomes. Table 1 gives the round per logic. The population both checkers see is the benchmarks for which the solver produced a proof of the empty clause: CVC5 answers unsatisfiable on 20 031 benchmarks but finishes printing the proof for 19 189 of them, and VERiT on 19 188 and 19 156 — the difference is the solver being killed mid-print, and counting it against the checkers would be counting a proof that does not exist. CARCARA accepts 19 188 and 18 680 of those proofs, and the checker validates 17 546 and 16 928. Of the proofs the checker does not validate, 1 627 and 1 916 are timeouts, none exhausts its memory on either solver, 0 and 10 have a trusted step, and 15 CVC5 proofs end in an error, all of one kind, which Section 2.4 takes apart. VERiT ends in no error at all, and CVC5 leaves no proof with a trusted step. The checker’s columns are independent verdicts on the same benchmark, not a partition: the checker validates more QF_UFLIA VERiT proofs than CARCARA accepts, because CARCARA reports a proof with a `lia_generic` step holey while the checker closes that step with `omega`. The picture per logic is uneven in an informative way: the quantified logics are all but complete (every UF proof but one, and 99.9% of UFLIA, for CVC5) and QF_LIA is at 94%, while QF_UF, whose proofs are an order of magnitude larger, is at 72% and 74%, and QF_IDL, whose bounded-model-checking proofs are large too, at 69% and 47%.

Why CARCARA does not check every proof. On this population the question has a sharp answer for CVC5: CARCARA accepts 19 188 of the 19 189 proofs, and the one it does not is a rejected `equiv_simplify` step. The 842 benchmarks that separate *unsat* from *proof* are not CARCARA’s — the solver answered within its 120 s internal limit and was killed by the 150 s external one while printing, leaving an empty file.

For VERIT the gap is 476 proofs, and it splits four ways: 261 are holey, all VERIT’s oracle rule `lia_generic`, which carries no certificate and which CARCARA does not check either; 150 exceed CARCARA’s own 300s limit, on proofs of a median 57 MB, a quarter of them above 238 MB; 37 exhaust the 12 GB during the check or the elaboration; and 28 are proofs with a step CARCARA’s own checker rejects (`and_simplify` and `minus_simplify`).

Two CARCARA defects account for the 868 proofs the gap lost over the two rounds before this one, and both are worth stating because each was mistaken for something else first.

The larger, 650 proofs, was a *parser* defect: the `--allow-int-real-subtyping` flag was consulted for the operators whose argument sorts are fixed and numeric (`+`, `*`, `/`, the order comparisons) but not for the positions where the sorts are merely required to agree — the arguments of `=` and `distinct`, the branches of an `ite`, and the arguments of an uninterpreted function. VERIT writes an integer literal on one side of an equality with a real-sorted term, so those proofs were rejected before checking began with *sort error: expected ‘Real’, got ‘Int’*, which is why the affected benchmarks are exactly the real-arithmetic families: 412 in QF_UFLRA (409 `mathsat`, 3 `FFT`), 189 in QF_LRA (95 `meti-tarski`, 84 `LassoRanker`, 9 `miplib`) and 49 in QF_RDL (47 `sal`). The same defect had appeared in an earlier three-logic round, on 194 QF_LRA proofs, where it had been mistaken for `lia_generic` residue. The fix is confined to those three polymorphic positions, so that `(div 1.0 2.0)` is still rejected, and the affected families now check *valid*, not merely *parse*.

The smaller, 190 proofs, was `ac_simp`’s checker ignoring the step’s `:premises`. VERIT emits the rule with premises when the flattening of a subterm was derived earlier — notably a rewrite under a binder, packaged as a `bind` subproof — so the conclusion is congruence over those equalities and not the structural flattening of its left-hand side, which is the only reading the checker had. The cost was not a checking gap but a lost proof, since elaboration validates as it goes: the affected benchmarks came back with neither an elaborated proof nor a verdict, and with the diagnosis past the runner’s byte cap, so that only 13 of the 190 could be attributed to the rule at all until the runner learned to lift the attribution line out first. The elaborator had carried a premise-aware decomposition all along, reached only for the steps whose premises are redundant, which is why the 11k proofs that use `ac_simp` were otherwise unaffected. The checker now reads the premises as sub-rewrites and mirrors the elaborator’s two routes: the structural reading first, and, failing that, normalizing both sides to a common form. That second route is the general fallback, and it is needed for more than the premises: VERIT also writes *premise-free* conclusions that are simply under-flattened, a nested `(and (and A B) C)` left alone deep inside a term whose normal form is `(and A B C)`. Reading the rule as “the two sides are equal modulo flattening and idempotence” accepts both shapes, is sound, and is what the elaborator had always computed; the elaborator decomposes either into per-layer `semilattice_simp` steps glued by `cong` and `trans`, which the checker already reconstructs, so no lean-smt change was needed. Of the 190 —

175 QF_UFIDL (`pete2`, `pete`, `uclid`) and 15 QF_LIA — all now check, and the nine QF_LIA `calypto` proofs among them are holey only for their `lia_generic` steps.

What survives is the 27 `and_simplify` rejections, on the same 24 benchmarks as the previous round plus three that could not be checked at all before. These are a VERiT-side shape the rule does not admit: the step’s left-hand side is not a conjunction, as in `(not (= (of_bool$ false) (of_bool$ true)))`, so the checker refuses it before any simplification is attempted.

Against CARCARA. Figure 2 plots, per proof, CARCARA’s check of the solver’s proof against the elaboration plus the checker’s check of the elaborated one. On the benchmarks *both* validate — 13 445 for CVC5 and 10 602 for VERiT — the checker is a median of 169× and 425× slower (geometric mean 146× and 386×; ninetieth percentile 406× and 888×). Elaboration is not what costs: comparing the two checks alone moves the medians to 167× and 423×. The factor varies by an order of magnitude across logics (last column of Table 1), and it is smallest exactly where the proofs are largest — 84× in QF_UF against 348× in UF — because the checker pays a fixed five-second cost to load its library that dominates on the small proofs of the quantified logics, while CARCARA, at a median of 0.04 s (CVC5) and 0.01 s (VERiT) over the round, is nearly free there. In QF_UF the medians are 0.79 s against 39 s and 0.14 s against 36 s, and the timeouts sit above proofs CARCARA disposes of in a second or two. Over the round CARCARA spends 9.6 and 8.9 CPU hours and the checker 211 and 218, against the previous round’s 212 and 223 on all but a handful of the same proofs. That is the price of a proof term the kernel accepts, and it makes the interesting question not whether the checker competes with a dedicated one but how far up the size distribution it reaches: below 10 000 elaborated steps the checker times out on 0.1% of CVC5 proofs and 0.7% of VERiT ones, between 20 000 and 50 000 on 31% and 64%, and above 50 000 on 90% and 95%. Those bands are where the reflective resolution checker of Section 2.2 told, and they have not moved since it landed: the middle one was 72% and 94% before it and 32% and 64% after. Nothing in this round addresses **resolution**, which is why the shape of the distribution is unchanged.

How the cost accumulates. Figure 3 plots, for each configuration, the cumulative solving and checking time against the number of benchmarks it validated, in the style of the LEAN-SMT paper. The checker follows CARCARA closely over the first thousands of benchmarks — the small proofs of the quantified logics, where both are dominated by the solving — and then separates in slope and in where it stops. Within the whole budget CARCARA needs for every proof it accepts, solving included (26.8 CPU hours for CVC5, 18.1 for VERiT), the checker validates the first 13 427 and 12 036 of them.

What is left. Rule coverage is complete on this corpus and the residue is at its floor. CVC5 leaves *no* proof with a trusted step, and VERiT leaves 10, every one of them `lia_generic` — the oracle rule that carries no certificate, which CARCARA does not check either and which the checker closes with `omega` only

Table 1. The SMT-LIB round, 23 328 benchmarks per solver. *proof* counts the benchmarks for which the solver produced a proof of the empty clause — it answered unsatisfiable *and* finished printing the proof within its limits, which is the population the two checkers see; CARCARA the proofs it accepts; then the checker’s outcomes — proofs validated with no trusted step, timeouts, runs that exhausted its 10 GB, and proofs with a trusted step (*holey*). The checker’s columns are independent verdicts, so they do not sum to CARCARA’s. The last column is the median factor by which the checker, elaboration included, is slower than CARCARA on the benchmarks *both* validate.

solver	logic	bench.	proof	CARCARA	LEAN-SMT				
					valid	t/o	mem	holey	slower
CVC5	QF_UF	4 361	4 295	4 295	3 106	1 189	0	0	84×
	QF_IDL	917	473	473	326	147	0	0	158×
	QF_RDL	113	73	73	39	34	0	0	120×
	QF_LIA	4 748	2 510	2 510	2 361	136	0	0	151×
	QF_LRA	703	526	526	428	98	0	0	86×
	QF_UFIDL	422	38	38	30	8	0	0	2×
	QF_UFLIA	192	172	172	165	5	0	0	172×
	QF_UFLRA	511	441	441	440	1	0	0	16×
	UF	3 505	3 155	3 155	3 154	1	0	0	348×
	UFIDL	53	53	53	53	0	0	0	176×
	UFLIA	7 793	7 443	7 442	7 434	8	0	0	160×
	UFLRA	10	10	10	10	0	0	0	—
total		23 328	19 189	19 188	17 546	1 627	0	0	169×
VERiT	QF_UF	4 361	4 187	4 173	3 068	1 105	0	0	372×
	QF_IDL	917	635	539	257	281	0	0	100×
	QF_RDL	113	86	76	41	35	0	0	192×
	QF_LIA	4 748	2 603	2 325	2 299	147	0	10	618×
	QF_LRA	703	574	573	450	123	0	0	172×
	QF_UFIDL	422	270	265	72	192	0	0	545×
	QF_UFLIA	192	173	130	173	0	0	0	516×
	QF_UFLRA	511	415	415	383	32	0	0	60×
	UF	3 505	2 897	2 881	2 881	0	0	0	449×
	UFIDL	53	51	51	51	0	0	0	485×
	UFLIA	7 793	7 255	7 242	7 243	1	0	0	430×
	UFLRA	10	10	10	10	0	0	0	—
total		23 328	19 156	18 680	16 928	1 916	0	10	425×

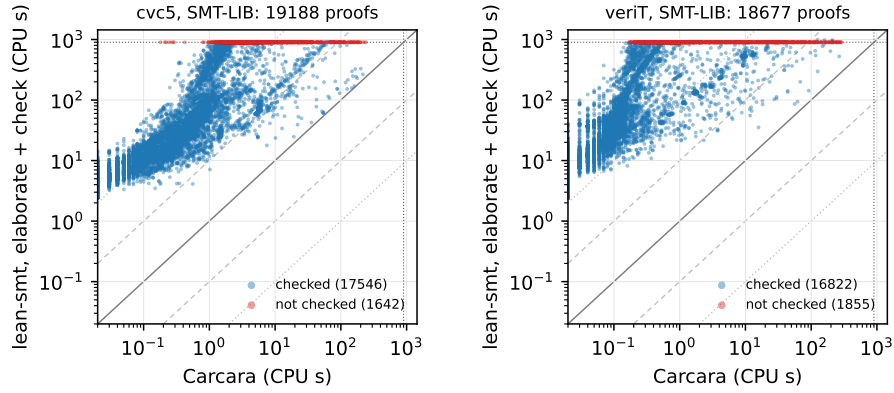


Fig. 2. CARCARA’s check of the solver’s proof against the elaboration plus LEAN-SMT’s check, on the SMT-LIB proofs CARCARA accepts, CPU seconds. Red points are proofs the checker did not finish within its limit; the grey lines are equality and factors of ten and a hundred.

when the goal is in that tactic’s fragment. VERIT ends in no error at all, and both figures are exactly what the previous round left: this round removed three rule reconstructors and narrowed a fourth (Section 2.2), and no step anywhere in the corpus became trusted as a result. CVC5’s 15 errors are the higher-order group of Section 2.4, the one gap left open deliberately. The rest is scale: the timeouts, 1 627 and 1 916 against the previous round’s 1 635 and 1 926 and the 2 259 and 2 729 before the reflective resolution checker. What separates this round from the last is where the time inside a proof goes, not how many proofs are checked; the verdicts move by six and seven proofs, all of them solver timing against the 150 s kill.

2.2 Where the Time Goes, Rule by Rule

The $169\times$ and $425\times$ of the previous section are whole-proof figures, and they say nothing about which steps are expensive. Because the round checks the *elaborated* proof twice — once with CARCARA, once with the checker — the two can be compared step for step over the same population: the 18 153 benchmarks whose elaborated proof CARCARA accepts *and* the checker runs to the end. On those the two tools see the same proof, and their counts of a rule’s steps agree exactly, benchmark by benchmark, which is what makes a single step count meaningful in the figure. Figure 4 draws, for the rules that carry the most total time, the distribution over benchmarks of the mean time per step of that rule.

The two solvers are pooled: their per-rule distributions are close enough that separating them says nothing the whole-proof figures did not already say. The distributions are wide but the ordering is stable, and three things stand out. First, the gap is not uniform: it runs from under $300\times$ on *rare_rewrite*, where the checker replays a rewrite it can name, to four orders of magnitude on the small clausal rules — $4\,600\times$ on *or_neg* and $5\,100\times$ on *and* — where CARCARA

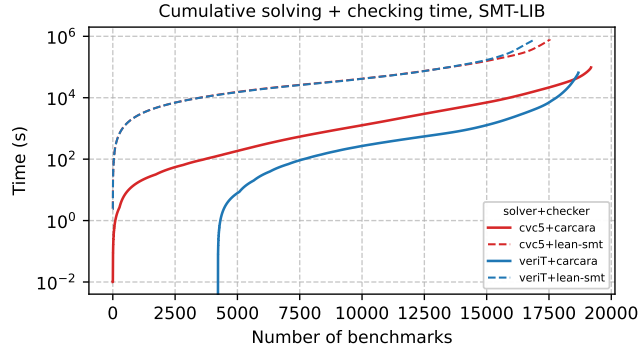


Fig. 3. Cumulative solving and checking time against the number of SMT-LIB benchmarks whose proof the configuration validated. CARCARA validates 19 188 CVC5 proofs in 26.8 CPU hours and 18 680 VERiT proofs in 18.1; the checker validates 17 546 and 16 928 in 219 and 223.

compares two arrays and the checker builds and kernel-checks a term. Those rules are cheap per step for both, but they are the most numerous by an order of magnitude (24M `cong` and 26M `resolution` steps over the population the figure draws), so the ratio there is what sets the whole-proof figure.

Second, one rule dominates the total: `resolution` is 61% of the checker’s 332 CPU hours, and `semilattice_simp` a further 11%. More than half of the number is one rule, which is why the remedy pursued for the timeouts was a reflective checker for exactly it. A clause is reified over a context of atoms — a literal is a natural number, a clause a list of them whose denotation unfolds definitionally to the proposition the step states, and a bitset on which one link of the chain is four machine-word operations — the whole chain is folded by the kernel in one evaluation, and a single soundness theorem, applied as a term, turns that evaluation into a proof of the stated clause. The pivots CARCARA writes are hints and never side conditions: resolving on a pivot that does not occur removes nothing, so a wrong hint can only surface as a failed containment at the end. Nothing is added to the trusted base — there is no `native_decide` here, and the soundness theorem depends only on `propext` and `Quot.sound`, inherited from the arithmetic lemmas it uses.

When that checker landed, the rule’s total fell from 87.4 to 36.8 CPU hours for CVC5 and from 73.3 to 27.3 for VERiT on the proofs both rounds validated, and the checker’s whole rule time from 132.4 to 77.8 and from 138.2 to 87.7 — the median proof gaining only 12% and 17%, too small to amortize the fixed cost of reifying the clauses, while the heaviest tenth of the proofs, which hold 87% of all `resolution` time, ran 2.6 and 2.9 times faster. Coverage follows the tail rather than the median, which is why 628 and 823 proofs crossed from timeout to validated while the whole-proof medians barely moved. This round leaves that machinery untouched and `resolution` sits where it was, at 202 hours over the drawn population; it is now the whole of the remaining problem. The

same containment check serves **contraction**, which falls to 0.19 and 0.33 of its previous cost. It does *not* serve every clausal rule: **or** and **weakening** were measured 3 to 4 times slower through it, because their old paths already did nothing — for **or** the premise’s proposition *is* the stated clause, and **weakening** is a single lemma application — so those keep their direct paths, as does a **contraction** whose premise is already its conclusion.

and was of another kind, and the previous round’s figure was what made us look. Its median then was 8.6ms, *above* **resolution**’s 5.2ms and higher than every rule in the figure but **poly_simp**, 30 000× **CARCARA**’s, and the rule was 14% of the checker’s time — the highest per-step cost of any high-volume rule, on a rule whose whole content is to name the *i*-th conjunct of a premise. That was an artefact of how the proof term was built rather than anything the rule demands. The reconstruction sends an *n*-ary **and** to a right-nested $p_0 \wedge (p_1 \wedge \dots)$, which is **andN** of the list of conjuncts only after *n* unfoldings; projecting the *i*-th conjunct *through* that list therefore charged every step for the whole conjunction — an *n*-element **List Prop** rebuilt per step, a **decide** of $i < n$ over it, *n* unfoldings of **andN** to reconcile the premise’s stated type, and *i* reductions of **List.getElem**. Projecting structurally instead costs nothing to reduce: the premise’s type is *already* headed by $\wedge$, so **And.right** applies with no unfolding at all, and the conjunct that comes out is the very term the conclusion was built from, so the kernel’s remaining check is a pointer comparison — the same thing **CARCARA** does with its hash-consed **Arc::ptr_eq**, arrived at from the other side. The same holds of **not_or** and of the CNF axioms, where **or_pos** collapses to a single node. The lesson generalizes past these rules: where the consumer’s type already *is* the connective, walk it; do not manufacture a list and index it. In this round the rule’s median is 1.4ms and its share of the checker’s time 3%.

reordering has left the figure altogether. It is an artefact of the elaboration rather than of any solver, and **CARCARA**’s own default pipeline removes such steps, recomputing the conclusions of the order-sensitive rules; ours had simply not run that pass. Since a clause is a set to this checker, those steps were pure cost — 2.7 and 1.3 CPU hours in an earlier round, and 107 of the 1 862 steps of one QF_UF proof — and running the pass removes them at no charge to the elaboration.

distinct_elim has left it too, and for a reason worth recording, because the diagnosis that looked obvious was wrong. The rule cost 5.17 CPU hours on **VERIT** in the previous round, over 11 751 steps in 1 982 proofs — 1.58 seconds a step, against 1.3 milliseconds for the 6 973 **CVC5** steps of the same rule. A **distinct** over *n* terms eliminates to the conjunction of its $\binom{n}{2}$ disequalities, and **VERIT** is free to write those conjuncts in any order and either orientation, so the checker had been aligning the stated conjunction against the one it builds and bridging the two with a congruence. On the worst proof that alignment took 288 seconds for a single step. The natural reading was that the kernel was reducing the congruence chain, and it was wrong: the **native** arm, which reduces none of it, paid 5.13 hours for the same steps. The cost was the alignment itself, in the reconstructor, and no amount of kernel work would have touched it.

What removes it is putting the canonical form on CARCARA’s side. Its `polyeq` pass now builds the elimination itself — the pairs in index order, each conjunct $\neg(x_i = x_j)$ with $i < j$ — and bridges to whatever the solver stated, so the checker sees one shape and closes the step with an `Eq.refl` ascribed to the stated equality. It refuses anything else, which is the whole reason the round’s trusted-step counts are worth reading twice: they did not move. The rule falls to 0.04 CPU hours, 12 milliseconds a step, a factor of 132, and identically on both arms. The bridge is not free — VERIT’s `cong` rises from 8.78 to 9.99 hours on 3% more steps, and `eq_symmetric` and `trans` by less — so about 1.35 of the 5.13 hours comes back. The net is still 3.8 hours, and the lesson is the general one this pipeline keeps teaching: a canonical form the producer can establish once is cheaper than an alignment the consumer must redo at every step.

Third, the structural AC rules that replaced `aci_simp` (Section 1.4) are cheap on both sides: `semilattice_simp` at a median of 4.8 ms and `poly_simp` at 11 ms for the checker, with ratios of $970\times$ and $760\times$, better than the clausal rules they are glued to. Splitting `aci_simp` into rules with a normal form did not cost checking time.

The arithmetic rules are where the round’s second change lands, and it is smaller than the first. `la_generic` is 5% of the checker’s time and `poly_simp` a further 3%, and both run through the same reflective polynomial normalizer, whose cost the kernel pays: on a profiled QF_LRA step 93% of the kernel’s time was the one `decide` that normalizer hands it, and 83% of *that* was the merge sort inside it. That merge compared monomials by asking $<$ and then $=$ over the same two variable lists, each through a `Decidable` instance chain that the kernel has to unfold. Keying the variable order by a `Nat` and deciding it in one pass of kernel-accelerated primitives makes the merge 1.7 to 2.5 times cheaper in isolation. Across the corpus it is worth $1.12\times$ on `la_generic` for both solvers — 4.83 to 4.31 CPU hours for CVC5 and 13.79 to 12.28 for VERIT — and $1.06\times$ on `poly_simp`. The gap between the isolated figure and the corpus one is the step size: the proof it was profiled on has `la_generic` steps of some 176 monomials, where the merge dominates, while the corpus median is far smaller and the fixed costs — flattening, the chain of `sum_ub` lemmas, the denotation bridging the reified form back to the stated goal — are most of what a step pays. The `native` arm is the control that says the attribution is right: it evaluates the same normalizer as compiled code and moved by 1% and 0%.

2.3 The Sledgehammer Set

The corpus. The 5000 SMT2 problems of the “Seventeen provers under the hammer” baseline set, the quantified Isabelle goals the LEAN-SMT paper evaluates on, with the same limits, and with CVC5 given the instantiation options that paper used. These proofs are two to three orders of magnitude smaller than the SMT-LIB ones: a median of 59 elaborated steps for CVC5 and 62 for VERIT, against tens of thousands in QF_UF.

Outcomes. Table 2 gives the round, and on this corpus the checker has caught up with CARCARA exactly. It validates *every* one of the 2869 CVC5 proofs CARCARA

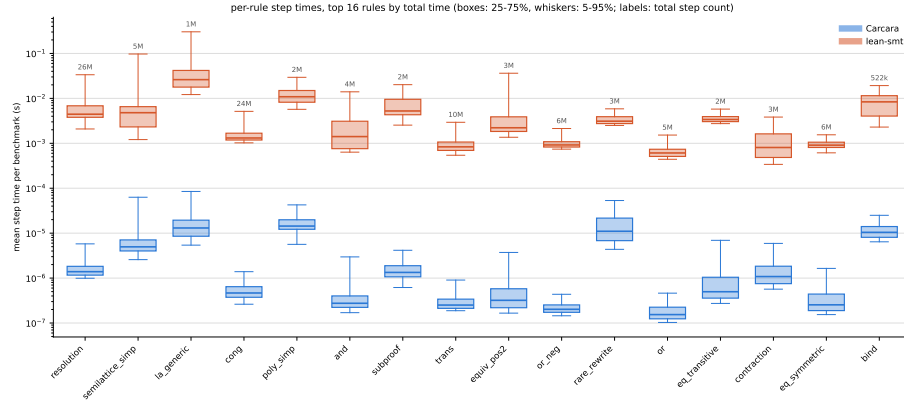


Fig. 4. Mean time per step of each rule, over the 18 153 benchmarks whose elaborated proof CARCARA accepts *and* the checker runs to the end, for the sixteen rules with the largest total time, pooling the two solvers. The two series share an x position. Boxes are the quartiles and whiskers the 5th and 95th percentiles; the label above each group is the rule’s step count over that population, which is the same number for both tools — they are counting the steps of the same proofs. Note the log scale: CARCARA sits three to four decades below LEAN-SMT throughout.

accepts, and 2592 of the 2600 VERiT ones (99.7%). Neither solver leaves a timeout or a memory failure. What limits the round is the solver — 2870 and 2681 of the 5000 problems are proved unsatisfiable at all — and, for VERiT, CARCARA’s own check, which rejects 81 proofs: 80 are a `and_simplify` step its checker does not accept, the same defect that rejects 102 VERiT proofs in the SMT-LIB round, and one is a `forall_inst`. Three more VERiT proofs that CARCARA’s check accepts never reach the checker: its elaborator panics on them in the `polyeq` pass (“terms not equal”) and prints no proof, which is why the LEAN-SMT columns of the table account for 2597 of the 2600.

The round before this one on the same corpus had a visibly larger residue — 2858 and 2520 validated, 9 and 51 proofs with a trusted step, three timeouts and 21 VERiT runs that exhausted memory — and the difference is what the rounds in between fixed, with nothing going the other way: no proof that validated then fails now. For CVC5, 8 of its 9 were a `bind` step the kernel rejected over VERiT-style anchors that both keep a variable and substitute it, and CARCARA’s `polyeq` now closes those subproofs by `ref1`; the ninth is a genuine `hole`, which CARCARA reports holey too, so the two checkers now agree on this corpus proof for proof. For VERiT, 21 errors and three timeouts became validations, and the trusted steps fell from 88 to 15, of which the 76 `onpoint` became 3 — the general handler following CARCARA’s `extract_points` — and the 9 `qnt_simplify` became none. CARCARA’s own rejections fell from 83 to 81, the three that went being sort errors from the integer/real printer defect.

Against CARCARA. Figure 5 is the comparison of Figure 2 on this corpus. CARCARA checks and elaborates in a median of 0.02 s and 0.01 s; the checker’s median

Table 2. The Sledgehammer round, 5 000 problems per solver; the columns of Table 1, with the CPU hours each checker spends on the proofs it validates (elaboration included for LEAN-SMT) and the median slowdown on the proofs both validate.

solver	proof	CARCARA	LEAN-SMT				CARCARA h	LEAN-SMT h	slower
			valid	t/o	mem	holey			
CVC5	2 870	2 869	2 869	0	0	1	0.02	4.2	239×
VERiT	2 681	2 600	2 592	0	0	5	0.01	3.8	484×

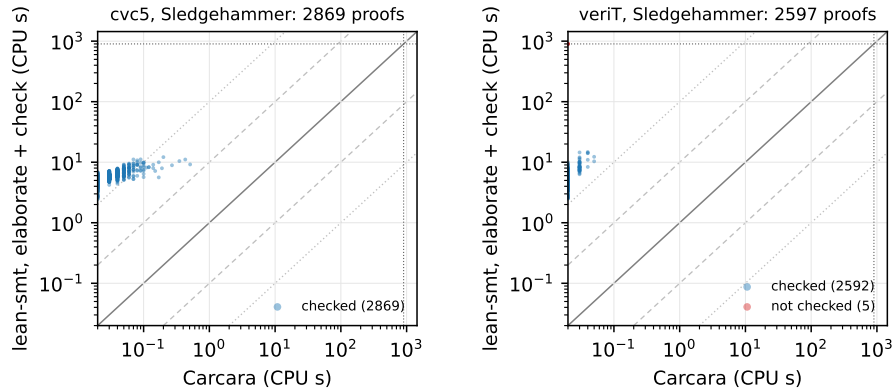


Fig. 5. CARCARA’s check against the elaboration plus LEAN-SMT’s check on the Sledgehammer proofs, CPU seconds. The band is the fixed cost of loading the library.

of 5.1s is almost entirely the cost of loading the library, not of checking, which is why the points form a horizontal band. On the 2869 and 2592 proofs both validate the checker is a median of 239× and 484× slower — the same factors as in the SMT-LIB round, for a different reason: there they measure a kernel replaying large derivations, here a fixed startup cost against a checker that finishes in hundredths of a second. Over the round the checker costs 4.2 and 3.8 CPU hours, against 72.2 and 92.8 hours of solving. The proof that the fixed cost dominates is the `native` arm: it validates the same 2869 and 2592 proofs at a median ratio of 0.95, against 0.93 in the SMT-LIB round, because on a proof of sixty steps there is almost nothing for it to evaluate — on the smoke benchmark, 10 of the checker’s 961 milliseconds were the kernel’s.

How the cost accumulates. Figure 6 is the cumulative plot for this corpus, and it looks different in an instructive way. Restricted to the proofs each configuration validates, solving is cheap — a median of a third of a second for CVC5 and a tenth for VERiT — so the curves are dominated by what comes after it. CARCARA disposes of all 2869 CVC5 proofs in 1.41 CPU hours including the solving, and of 2600 VERiT ones in 1.21; the checker takes 5.7 and 4.9 for 2869 and 2592 of them, four times as much for the same proofs or all but eight of them. The checker’s curve is close to a straight line, because on proofs this small its cost is

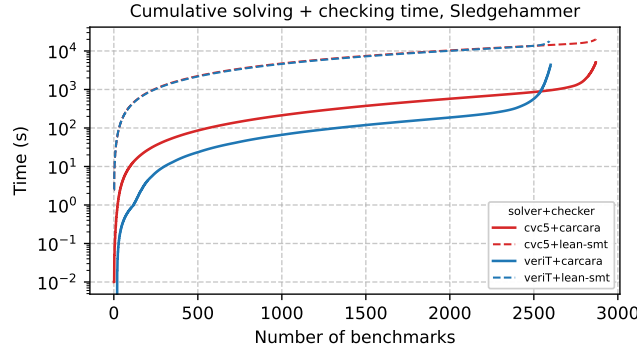


Fig. 6. Cumulative solving and checking time against the number of Sledgehammer benchmarks whose proof the configuration validated. The LEAN-SMT curves are nearly straight.

the fixed one of loading the library rather than anything about the derivation. Within CARCARA’s whole budget the checker validates the first 1115 (CVC5) and 971 (VERIT).

What is left. Nine proofs in 5470 — the 2870 CVC5 and 2600 VERIT proofs CARCARA does not reject. Three are the VERIT proofs its elaborator drops before the checker sees them, the same panic on all three. The six the checker does see are each a distinct question rather than an instance of a pattern. CVC5 leaves one, the genuine `hole`. VERIT leaves five: two `onpoint` steps whose guard the traversal still does not find — one where the conjunct sits under an application of a term-level function symbol, one where the body is a bare Isabelle `thesis` variable with no guard at all; two proofs whose `eq_transitive`, `eq_congruent` and `la_generic` steps are stated over `let`-bound numerals, where the chain matching compares the `let` spellings rather than what they denote; and one `bind` the kernel rejects with an application type mismatch, the last survivor of the group that made up eight of CVC5’s nine in the previous round. The `let` group is the interesting one, because it is the same shape the SMT-LIB round meets as numeral matching in assumptions (Section 1.3): two readings of one term that agree on value and differ on spelling.

2.4 What the Proofs Revealed

Checking against a second, independent reading of the rules paid off in both directions. On the producer side: VERIT’s open named terms, its identity substitutions (`:= (?x S) ?x`) in every `bind`, its fresh names colliding with inner binders, the eleven Sledgehammer proofs whose `bfun_elim` the core pass silently kept, CARCARA’s modus-ponens recipe, and the annihilator that `aci_simp` did not name. On the checker side: an `onpoint` guard refutation building `Or.inl` with an application builder that cannot infer the other disjunct, exposed only when the cluster reached the CVC5 UF benchmarks; the `Decidable` instance mismatch of Section 1.3; the quadratic construction of batch terms, caught by the

first batch sweep; the term-mode premise types; the hoisted definitional checks, which cost 49s on a `trans` step whose links all matched syntactically; and the polynomial normalizer’s insertion fold, the per-literal tightening that gave `omega` the whole local context, and the pairwise `distinct_elim` conversion, each a term whose size the kernel could not bear — and, later, the alignment that replaced that conversion, which was not the kernel’s cost at all but the reconstructor’s, and which only a canonical form on CARCARA’s side removed (Section 2.2). A rule-by-rule audit of the reconstructors against CARCARA’s checkers, done after the SMT-LIB round, closed the remaining differences a solver can produce — the four argument orientations of `cong` between two equalities, `distinct_elim` over Booleans, both forms of `la_tautology`, the empty clause from `(not true)`, and `lia_generic` by `omega`. Every finding is a regression test.

The failures the round leaves. Beyond the timeouts, 15 CVC5 runs end in an error and VERiT none at all. Two rounds ago there were 50 and 116, from four distinct defects of the checker’s front end rather than of any rule; all are now fixed and the rounds since confirm them.

Symbols with an apostrophe (22 CVC5 and 27 VERiT proofs, in UFLIA’s `tokeneer` and QF_IDL’s `sep`). CARCARA’s lexer admits `'` inside a simple symbol so that the variables its capture-avoiding substitutions rename (`x'`) round-trip, and its printer decided what to quote with the same predicate. A symbol read from a quoted `|status'|` was therefore printed back bare, which the SMT-LIB grammar does not allow, so the elaborated proof was rejected by CVC5’s parser — the checker’s own reader — with *Symbol ‘status’ not declared as a variable*. The printer now quotes any symbol containing an apostrophe; quoting is always safe, since `|s|` and `s` denote the same symbol. All 49 proofs now check.

Two quadratic costs in reading a large problem (the QF_LIA `Dartagnan` benchmarks, whose files reach 20 MB). Their assertions nest single-binding `lets` 117 000 deep, and the parser extended the binding environment by copying a hash map at every level, which is quadratic in the depth; persistent maps make it linear and the arena stage of one such problem falls from 150 to 22 s. Behind it, the check deciding whether a reconstructed term mentions a variable of an enclosing scope — the test that `lets` closed terms be cached across scopes — used a traversal that memoizes only within a single call, so it was linear per term and quadratic over an assertion of 700 000 nodes; memoizing per node in the reconstruction state took one proof that had not finished reconstructing its six assertions in 17 CPU minutes to a complete check in 209 s. Both benchmarks that had timed out over this in the previous round now validate, and faster than they did two rounds ago.

Problems with more declarations than the interpreter’s stack (5 CVC5 proofs, QF_LIA’s `SharedMemory` and `wireRouting`). Lean’s `withLocalDecls` nests one `withLocalDecl` per declaration, so introducing the 160 000 symbols and 280 000 assertions of those problems overflowed the interpreter’s stack after minutes of work. A flat variant that extends the local context in a loop and installs it once checks the 15 MB benchmark in 190 s; four of the five now check, and the fifth runs to the limit.

A *resolution loop* (2 of the 8 CVC5 memory exhaustions). Alethe’s resolution removes *every* occurrence of the pivot, so the chain re-resolves while the working clause still holds it. Against a tautological premise (`c1 l (not l)`), which CVC5 emits for `not_equiv2`-style splits, each pass erased the pivot and the premise put it straight back: two million iterations before the memory cap. The loop now stops when the premise contributes the pivot with the working clause’s own polarity.

The memory exhaustions that survived that fix had a separate cause, and it was not the one assumed: not a step too large but a quadratic *search*. A `cong` step aligns its premises with the argument positions of the two applications, and it did so by scanning the whole premise list for each argument. On a congruence over an `and` of 8 449 conjuncts — VERiT emits these on the `Dartagnan` benchmarks — that scan ran for over a minute and exhausted the 10 GB before the kernel saw anything. Indexing the premises by their two sides makes the alignment linear and the step costs 105 ms. With that and the parser fixes below, every memory exhaustion of the previous round is gone from both solvers.

An *instance lookup that threw* (40 VERiT proofs then, 30 in QF_UFIDL’s `pete2`, 8 in QF_IDL’s `RTCL`, 2 in `uclid`). The reconstruction of an `ite` over propositions needs a `Decidable` instance and searches for one before falling back to a compositional classical instance. On these deeply nested ground chains over integer differences the search threw its own *failed to synthesize* rather than reporting no instance, so the fallback never ran; and since this happens while the problem’s assertions are being reconstructed, before the first step, it escaped the per-step guard that turns a failed reconstruction into a trusted step and killed the run. A throw from the lookup is now a fall-through. The 40 proofs, at a median of 61 600 elaborated steps, belong with the timeouts, and that is what they now are.

One group is not fixed. 15 CVC5 proofs, 13 of them in QF_LIA’s `incrementalScheduling` and 2 in QF_UFLIA, need higher-order reasoning the checker does not have: CVC5 keeps a parameterised `define-fun` as an uninterpreted symbol plus an assumption equating it to a `lambda`, rewrites inside that `lambda` under a `bind`, and applies the result with `ho_cong`. The checker macro-expands definitions, so the unapplied symbol means nothing to it, and it has no `ho_cong`; these are the only proofs in the round that use the rule. Supporting them means binding `define-funs` as functions rather than expanding them.

3 Per-Step Against One-Term Submission

The per-step shape is the default, and the measurements above use it. This section records what led to that choice: the comparison with the one-term shape, in which the whole derivation is reconstructed as one term and checked in a single kernel call — the shape the CPC path uses — made on the three-logic SMT-LIB round that preceded the twelve-logic one (QF_UF, QF_LIA and QF_LRA, 9 812 benchmarks per solver) and on the Sledgehammer round. The other variant of the default shape, `native`, is the subject of Section 4.

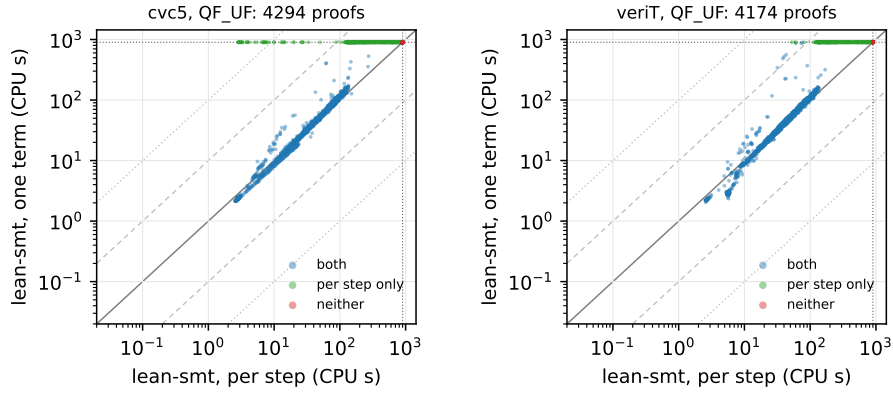


Fig. 7. The two kernel submission shapes on the same QF_UF proofs, CPU seconds. A point on a dotted border is a proof the corresponding shape did not check; the grey lines are equality and factors of ten and a hundred.

SMT-LIB. Figure 7 puts the two shapes side by side. Per-step checking dominates in every logic and for both solvers: it validates 1 603 proofs (CVC5) and 1 644 (VERiT) that the one-term shape does not, and there is not a single proof in the other direction. On the proofs both check it is also slightly cheaper. The one-term failures that are not timeouts split in two. About six hundred per solver are the single kernel call rejected for excessive memory, the cost of holding the whole derivation at once. The other thousand per solver were a defect of term mode found by this run: step by step, a premise is a hypothesis whose declared type is the clause the proof states, while in one term it is the proof itself, whose inferred type is only definitionally that clause, and the congruence reconstructor, reading a premise’s type, saw something else. A term-mode step now carries its stated type; the memory failures remain, and the measurement that led us to keep per-step checking as the default (Section 1.2) holds at corpus scale.

Sledgehammer. On proofs this size the two shapes are nearly indistinguishable — 2 725 against 2 858 for CVC5, 2 508 against 2 520 for VERiT, at the same median cost (Figure 9) — which is the other side of the SMT-LIB finding: the one-term shape loses only when the derivation is too large to hold at once. The per-step column here is that earlier round’s, not the one Section 2.3 reports: the one-term arm has not been run since, so the comparison is drawn as it was measured. Of the one-term failures, 133 (CVC5) and 12 (VERiT) were the term-mode premise-type defect, since fixed; with it the two columns essentially coincide, and every remaining difference is a proof the per-step arm also fails.

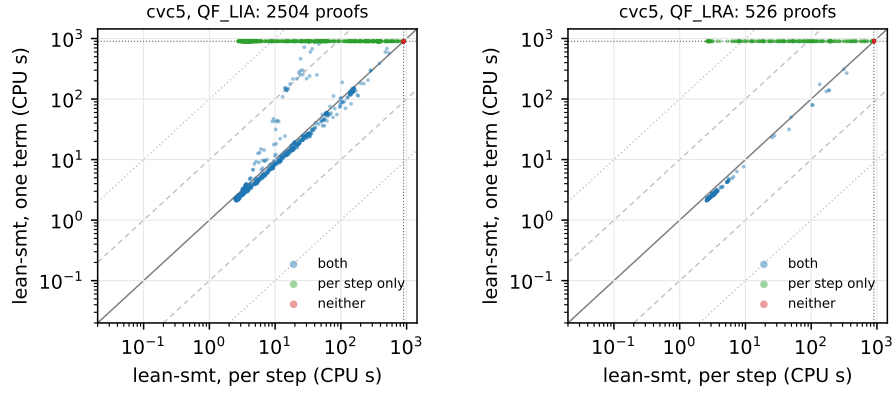


Fig. 8. The same comparison on cvc5’s arithmetic proofs, QF_LIA (left) and QF_LRA (right). The proofs are smaller, so both shapes reach further, and the one-term shape still loses 812 and 217 benchmarks.

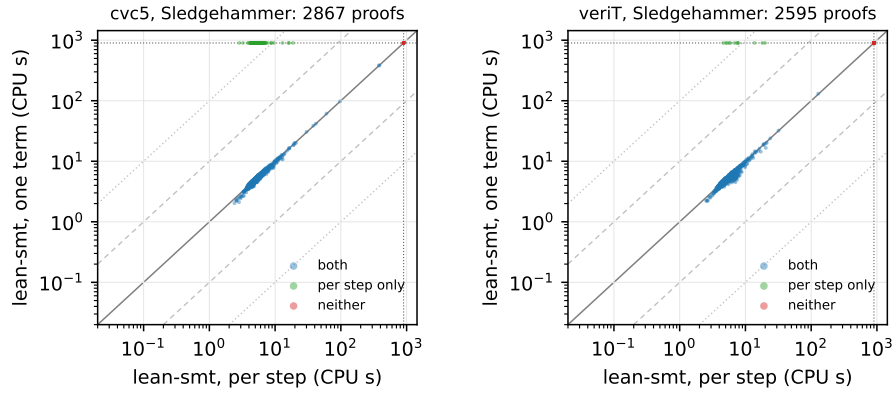


Fig. 9. The two submission shapes on the Sledgehammer proofs, CPU seconds. The band along the diagonal is the fixed cost of loading the library.

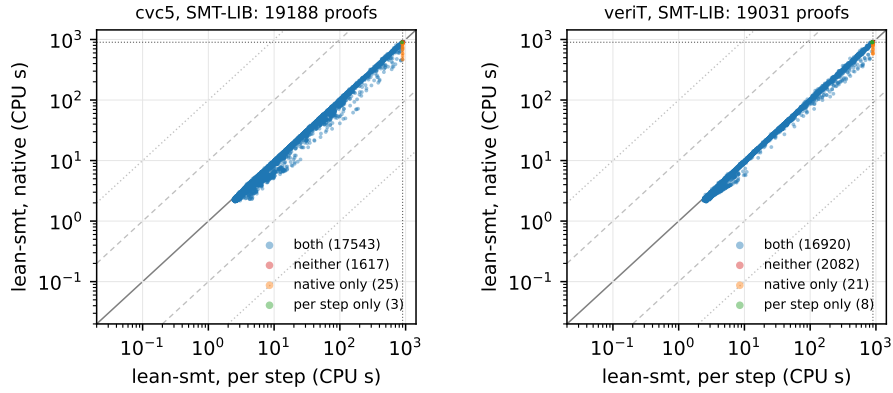


Fig. 10. The per-step shape against its **native** variant on the same SMT-LIB proofs, CPU seconds. A point on a dotted border is a proof the corresponding arm did not validate; the grey lines are equality and factors of ten and a hundred. The cloud sits just below the diagonal, and the points either arm loses alone are all against the limit.

4 The native Variant

The reconstructors have a second mode. Under **native**, the polynomial normalizer behind the arithmetic rules and the decision procedure behind the decidable side conditions — the sign facts of `la_generic`, the nonzero coefficient of `poly_simp_rel`, and `evaluate`’s ground equalities — run as compiled Lean code, and their verdicts enter the proof through `nativeDecide` instead of being reduced by the kernel. It buys time and costs trust: Lean’s compiler and runtime join the kernel and CVC5’s parser in the trusted base. Both sides are measured here on the twelve-logic SMT-LIB round, where the two arms ran on the same node over the same elaborated proof, under the same 900s limit and with the same per-step batching, so that every benchmark is a matched pair.

Outcomes. Table 3 gives the round per logic. **native** validates 22 more CVC5 proofs (17568 against 17546) and 13 more VERiT ones (16941 against 16928), a tenth of a percent of either population, and what gain there is is timeouts avoided: 1627 fall to 1605 and 1916 to 1903. Nothing else moves. Neither arm exhausts memory, the proofs with a trusted step are the same 0 and 10, and no proof’s trusted-step count differs between the arms.

Paired. Figure 10 puts the two arms on the same proof. Per logic the two columns of validated proofs are equal or differ by a handful, and the paired counts say why: 17543 (CVC5) and 16920 (VERiT) proofs are validated by both arms, 25 and 21 by **native** only, 3 and 8 by the plain arm only. Both asymmetries sit against the limit: every proof either arm wins alone is one the other left running at the 900s mark. What separates the arms is how close a proof already was to the timeout, not what either can check.

Table 3. The per-step shape against its **native** variant on the SMT-LIB round, over the benchmarks for which both arms ran: proofs validated with no trusted step and timeouts, per arm; then the paired verdicts — validated by *both* arms, by the plain arm only, by **native** only. The outcomes not shown are the same for both arms (no memory failure on either solver, 0 and 10 proofs with a trusted step). The last column is the median ratio of **native**’s CPU time to the plain arm’s, over the proofs both validate.

solver	logic	per step		native		paired			ratio
		valid	t/o	valid	t/o	both	step	nat.	
cvc5	QF_UF	3 106	1 189	3 108	1 187	3 103	3	5	0.988
	QF_IDL	326	147	328	145	326	0	2	0.964
	QF_RDL	39	34	41	32	39	0	2	0.827
	QF_LIA	2 361	136	2 371	126	2 361	0	10	0.846
	QF_LRA	428	98	433	93	428	0	5	0.857
	QF_UFIDL	30	8	30	8	30	0	0	0.922
	QF_UFLIA	165	5	166	4	165	0	1	0.878
	QF_UFLRA	440	1	440	1	440	0	0	0.936
	UF	3 154	1	3 154	1	3 154	0	0	0.931
	UFIDL	53	0	53	0	53	0	0	0.921
	UFLIA	7 434	8	7 434	8	7 434	0	0	0.915
	UFLRA	10	0	10	0	10	0	0	0.914
total		17 546	1 627	17 568	1 605	17 543	3	25	0.93
VERiT	QF_UF	3 068	1 105	3 070	1 103	3 067	1	3	0.985
	QF_IDL	257	281	262	276	255	2	7	0.980
	QF_RDL	41	35	42	34	41	0	1	0.928
	QF_LIA	2 299	147	2 295	151	2 294	5	1	0.873
	QF_LRA	450	123	459	114	450	0	9	0.871
	QF_UFIDL	72	192	72	192	72	0	0	0.984
	QF_UFLIA	173	0	173	0	173	0	0	0.951
	QF_UFLRA	383	32	383	32	383	0	0	1.001
	UF	2 881	0	2 881	0	2 881	0	0	0.915
	UFIDL	51	0	51	0	51	0	0	0.931
	UFLIA	7 243	1	7 243	1	7 243	0	0	0.930
	UFLRA	10	0	10	0	10	0	0	0.902
total		16 928	1 916	16 941	1 903	16 920	8	21	0.94

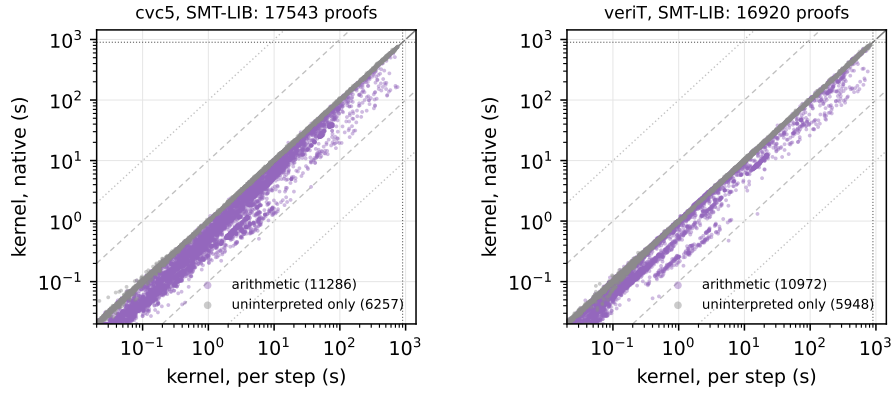


Fig. 11. The kernel’s own time in the two arms, on the proofs both validate, seconds; the proofs of a logic with arithmetic are drawn apart from those of QF_UF and UF. The saving is where the normalizer runs: the uninterpreted proofs lie on the diagonal, the arithmetic ones below it.

Where the time goes. On the proofs both arms validate, **native** is faster on 92.3% (CVC5) and 91.5% (VERiT) of them, and the median ratio of its CPU time to the plain arm’s is 0.93 and 0.94, with the middle half of the proofs between 0.88 and 0.97. Summed, 206.1 CPU hours become 198.4 and 213.5 become 209.0. Figure 11 isolates the kernel’s share of that: its median falls from 626 to 388ms per CVC5 proof and from 607 to 529ms per VERiT one, 111.4 CPU hours to 100.4 and 142.0 to 134.9, but it still accounts for 51% and 65% of each arm’s time against 54% and 67% before. The saving is a slice of the kernel’s work rather than most of it, because what the kernel spends its time on is the resolution chains and the congruences, which **native** does not touch. The per-logic medians of the last column of Table 3 locate the gain exactly: 0.83 to 0.86 in CVC5’s QF_LIA, QF_LRA and QF_RDL, where the normalizer runs on every arithmetic step, against 0.988 and 0.985 in QF_UF, where it never runs at all. The gap has narrowed slightly on the arithmetic logics since the previous round, which is the expected shadow of the normalizer’s monomial order being made cheaper for the kernel (Section 2.2): the plain arm moved towards **native**, not the other way.

What it is worth. The variant changes nothing about what is admitted without proof: over the 34 463 matched pairs both arms validate, the number of trusted steps is identical on every single one, and the holey population is the same. So **native** buys 6% of the median checking time and a tenth of a percent of the corpus, at the price of the compiler and the runtime in the trusted base. That is the wrong trade for a checker whose point is the kernel, and the plain per-step shape stays the default; the variant is kept for the one case its numbers do favour, a large arithmetic proof that will not check in the time available otherwise.

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